

- 3** Many food crops, such as potatoes and bananas, contain starch. The starch content can affect the taste and cooking properties of the food.

Potatoes can be boiled or baked. Potatoes with a high starch content are good for baking and those with a lower starch content are good for boiling.

It is suggested that King Edward and Russet potatoes are better for baking and Nicola and Purple Congo potatoes are better for boiling.

A student decided to test these suggestions.

A standard set of starch solutions was prepared and tested.

The table shows the results.

Starch concentration (%)	Colour of solution after addition of iodine solution
1.0	black
0.1	dark blue
0.01	pale blue
0.001	very pale blue
0.0001	pale brown

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- (a) Describe how the data in the table could have been obtained using a 1% starch solution.

(4)

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Potato variety	Colour
King Edward	black
Nicola	very pale blue
Purple Congo	pale blue
Russet	dark blue

(4)

(c) Suggest how this investigation could be adapted to produce quantitative data.

(2)

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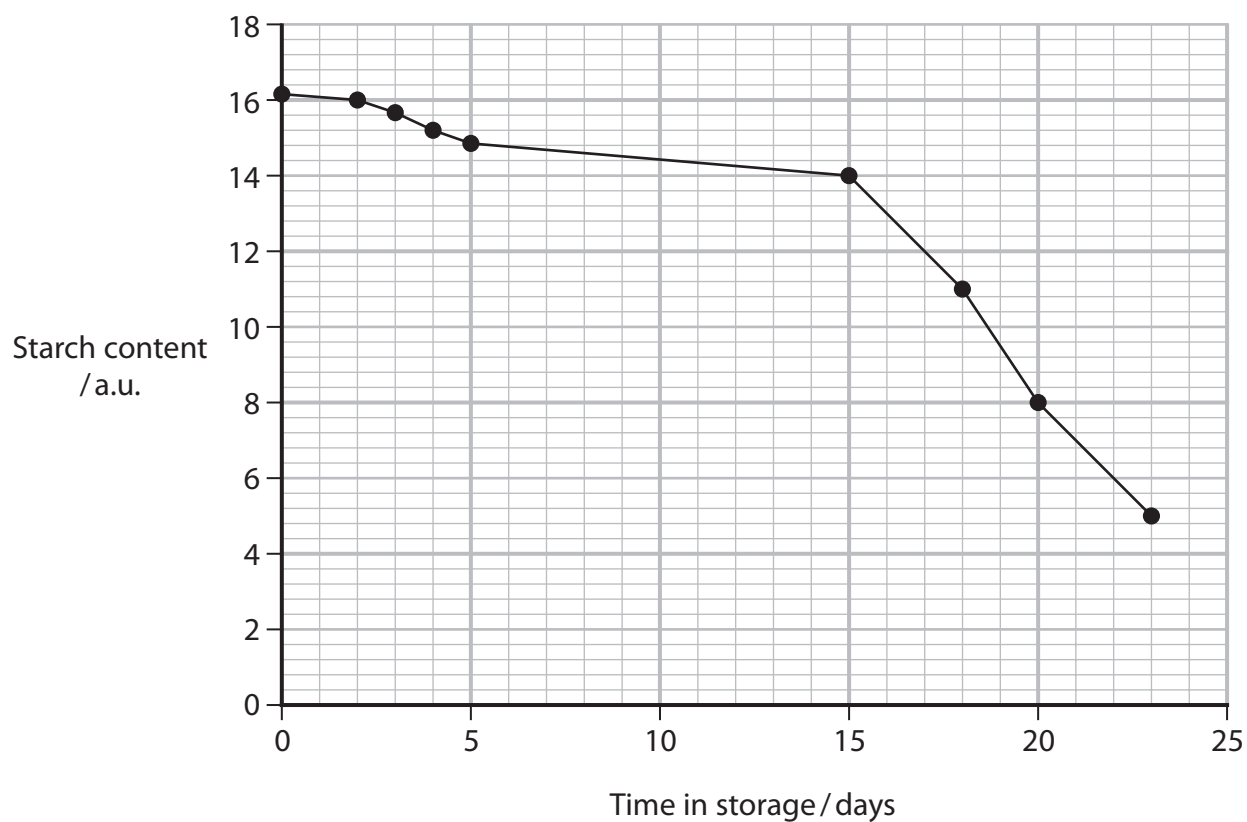
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- (d) It is suggested that the change in the taste and consistency of stored bananas is due to changes in their starch content.

An investigation of the content of starch in stored bananas produced the following results.



- (i) Calculate the rate of decrease in starch content from day 15 to day 23.

Include units in your answer.

(2)

Answer

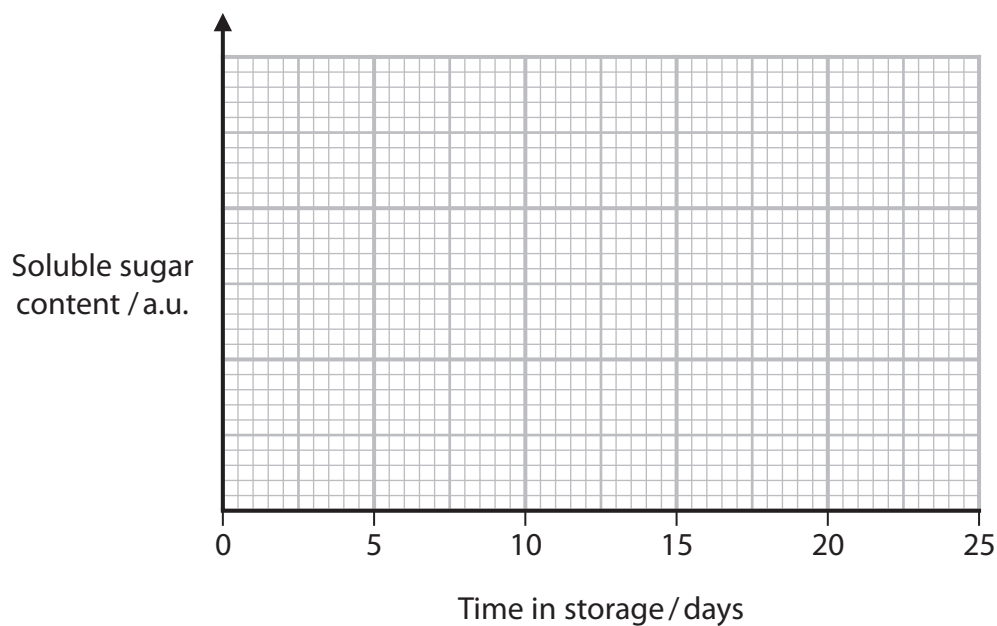


- (ii) Describe the changes in starch content during the 23 days of this investigation.

(2)

- (iii) Sketch a graph to show the soluble sugar content of these bananas during storage.

(2)



(Total for Question 3 = 16 marks)

TOTAL FOR PAPER = 50 MARKS

